CSE276C - Regression and Classification

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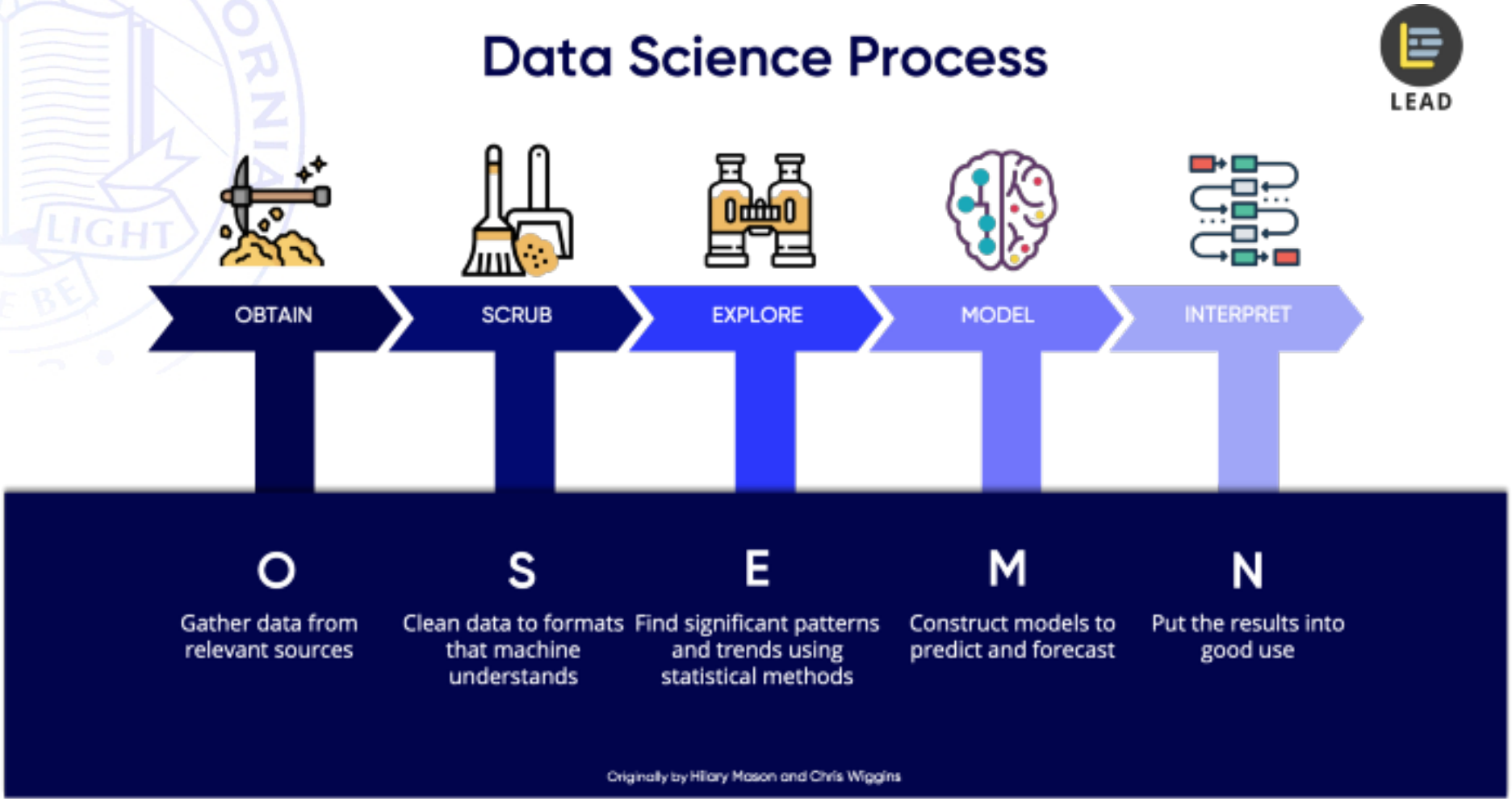
Computer Science and Engineering
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Introduction

- Data science is a big part of robotics
- Many aspects of robotics rely on data analysis
  - Recognition of objects
  - Adaptive Control
  - Clean-up of sensor information
  - ...

A general framework



Outline

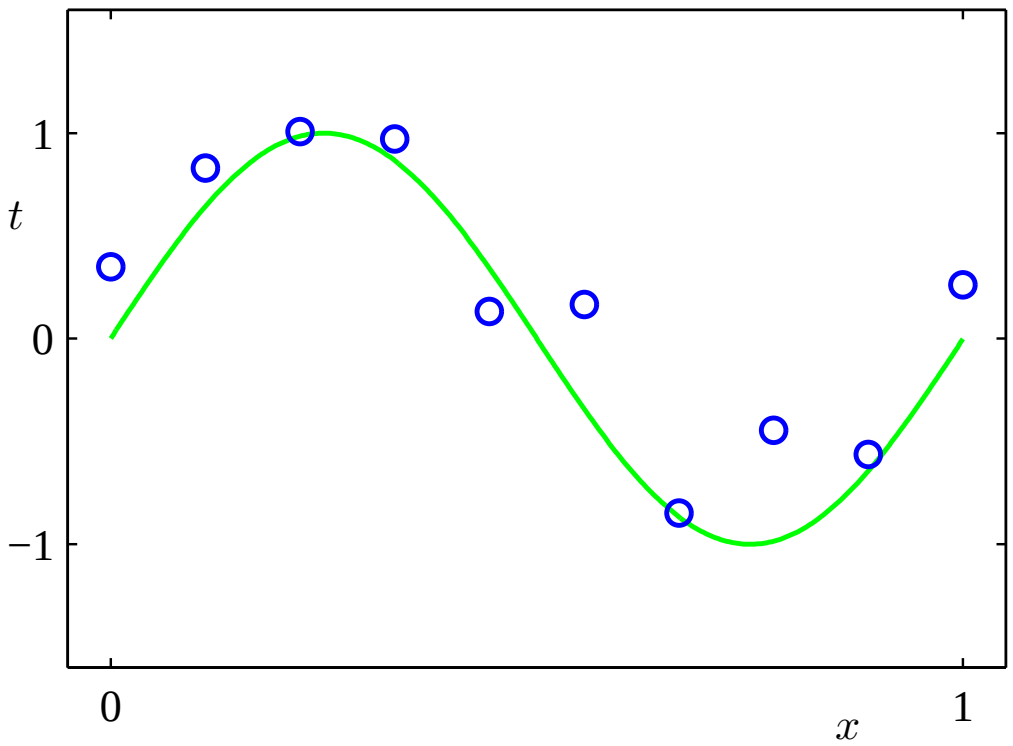
1. Introduction - Regression
2. Preliminaries
3. Linear Basis Function Models
4. Bayesian Linear Regression
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8. Linear Discriminant Functions
9. LSQ for Classification
10. Fisher's Discriminant Method
11. Perceptrons
12. Summary

Introduction

- The objective of regression is to enable prediction of a value based on modeling over a data-set X.
- Consider a set of D observations over a space
- How can we generate estimates for the future?
  - Battery time?
  - Time to completion?
  - Position of doors?

Introduction (2)

- Example



$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_m x^m = \sum_{i=0}^{m} w_i x^i$$

Introduction (3)

- In general the functions could be beyond simple polynomials
- The "components" $(\phi_i(x))$ are termed *basis functions*, i.e.

$$y(x, \mathbf{w}) = \sum_{i=0}^{m} w_i \phi_i(x) = \vec{w}^T \phi(\vec{x})$$

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Loss Function

- For optimization we need a penalty / loss function

$$L(t, y(x))$$

- Expected loss is then

$$E[L] = \int \int L(t, y(x)) p(x, t) dx dt$$

- For the squared loss function we have

$$E[L] = \int \int \{y(x) - t\}^2 p(x, t) dx dt$$

- Goal: choose y(x) to minimize expected loss (E[L])

Loss Function

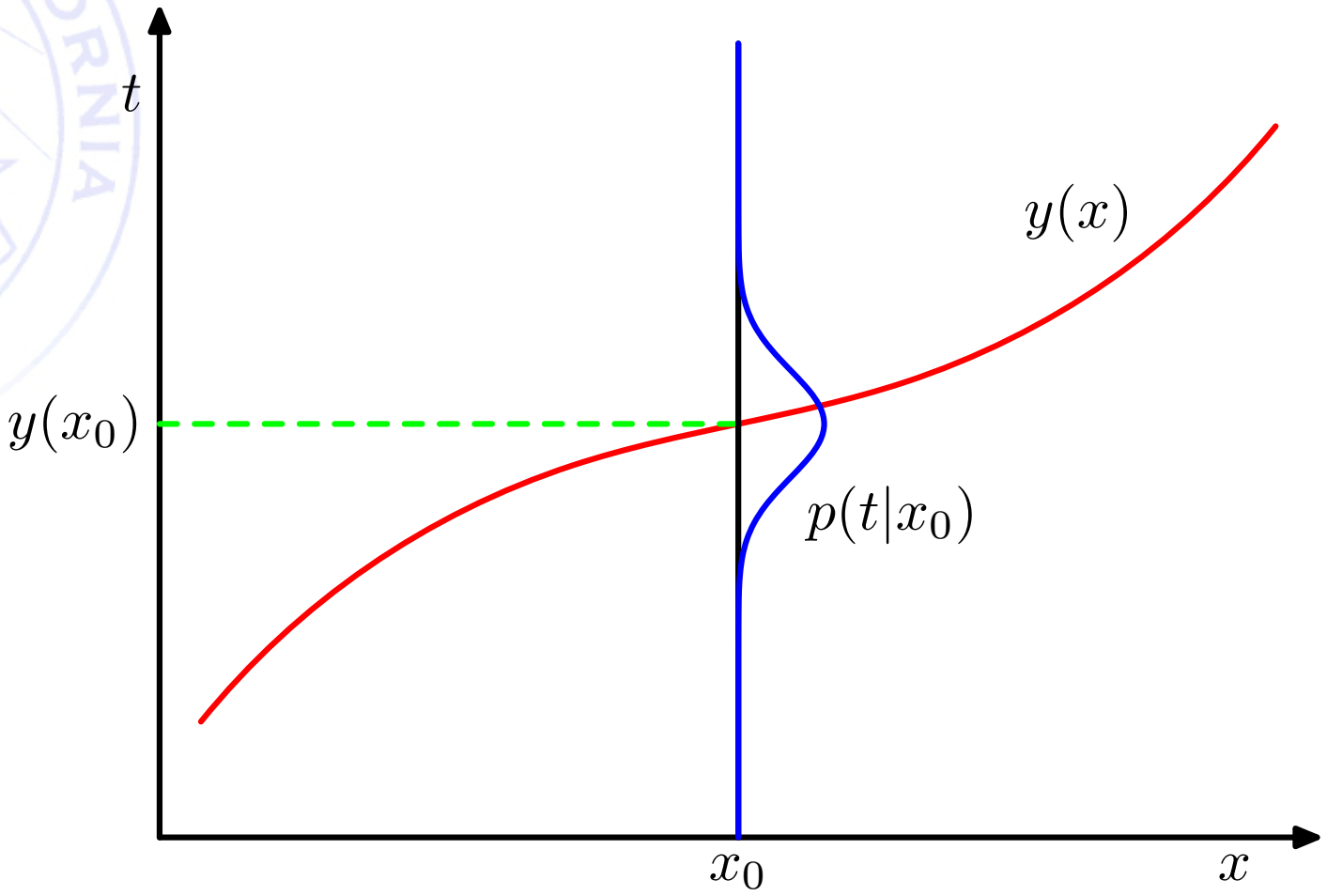
- Derivation of the extremum

$$\frac{\delta E[L]}{\delta y(x)} = 2 \int \{y(x) - t\} p(x,t) dt = 0$$

- Implies that

$$y(x) = \frac{\int t p(x,t) dt}{p(x)} = \int t p(t|x) dt = E[t|x]$$

Loss Function - Interpretation



Alternative

- Consider a small rewrite

$$\{y(x) - t\}^2 = \{y(x) - E[t|x] + E[t|x] - t\}^2$$

- The expected loss is then

$$E[L] = \int \{y(x) - E[t|x]\}^2 p(x) dx + \int \{E[t|x] - t\}^2 p(x) dx$$

Outline

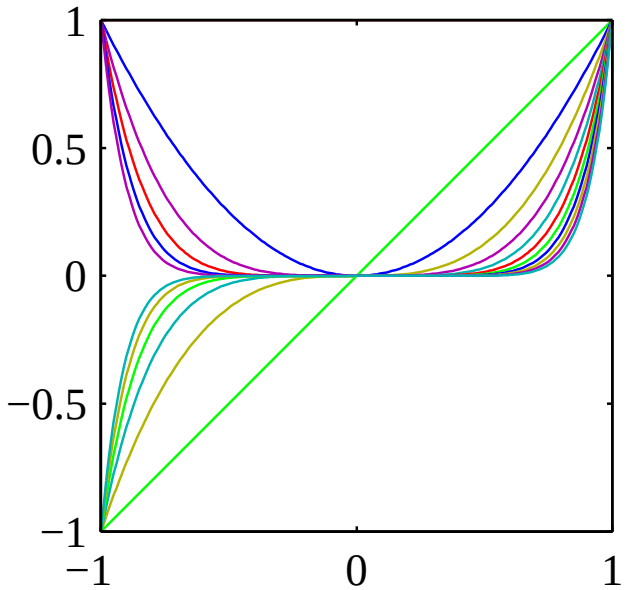
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Polynomial Basis Functions

Basic Definition:

$$\phi_i(x) = x^i$$

Global functions
Small change in x affects all of them

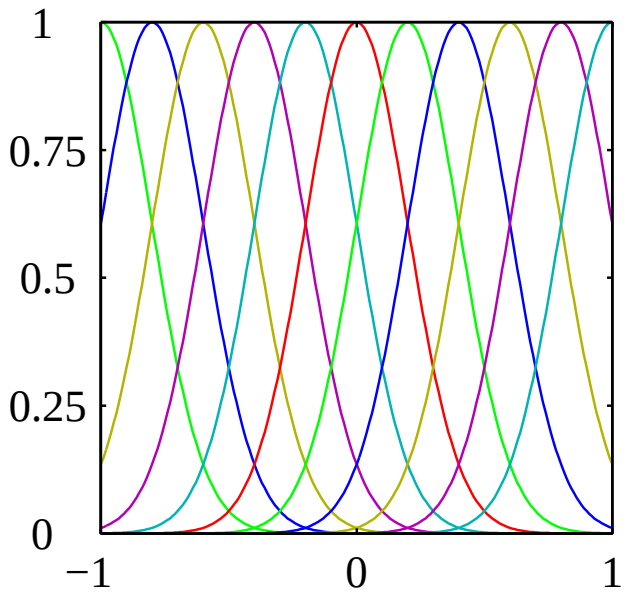


Gaussian Basis Functions

Basic Definition:

$$\phi_i(x) = e^{-\frac{(x-\mu_i)^2}{2s^2}}$$

A way to Gaussian mixtures, local impact
Not required to have probabilistic interpretation.
μ control position and s control scale



Sigmoid Basis Functions

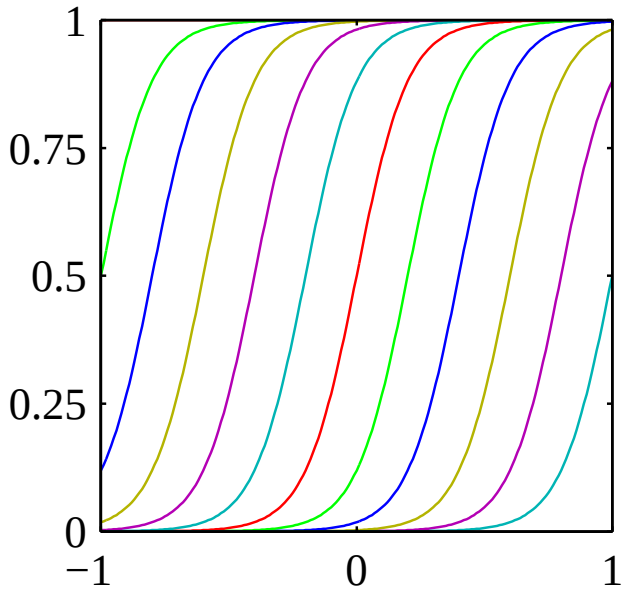
Basic Definition:

$$\phi_i(x) = \sigma\left(\frac{x - \mu_i}{s}\right)$$

where

$$\sigma(a) = \frac{1}{1 + e^{-a}}$$

μ controls location and s controls slope



Maximum Likelihood & Least Squares

- Assume observation from a deterministic function contaminated by Gaussian Noise

$$t = y(x, w) + \epsilon \quad p(\epsilon|\beta) = N(\epsilon|0, \beta^{-1})$$

the problem at hand is then

$$p(t|x, w, \beta) = N(t|y(x, w), \beta^{-1})$$

- From a series of observations we have the likelihood

$$p(\mathbf{t}|\mathbf{X}, w, \beta) = \prod_{i=1}^{N} N(t_i|w^T \phi(x_i), \beta^{-1})$$

Maximum Likelihood & Least Squares (2)

- This results in

$$\ln p(\mathbf{t}|\mathbf{w},\beta) = \frac{N}{2}\ln\beta - \frac{N}{2}\ln(2\pi) - \beta E_D(\mathbf{w})$$

- where

$$E_D(\mathbf{w}) = \frac{1}{2}\sum_{i=1}^{N}\{t_i - \mathbf{w}^T\phi(x_i)\}^2$$

  is the sum of squared errors

Maximum Likelihood & Least Squares (3)

- Computing the extrema yields:

$$\mathbf{w}_{ML} = \left(\Phi^T \Phi\right)^{-1} \Phi^T \mathbf{t}$$

- where

$$\Phi = \begin{pmatrix} \phi_0(x_1) & \phi_1(x_1) & \cdots & \phi_{M-1}(x_1) \\ \phi_0(x_1) & \phi_1(x_2) & \cdots & \phi_{M-1}(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(x_N) & \phi_1(x_N) & \cdots & \phi_{M-1}(x_N) \end{pmatrix}$$

Line Estimation

- Least square minimization:
  - Line equation: $y = ax + b$
  - Error in fit: $\sum_i (y_i - ax_i - b)^2$
  - Solution:

$$\begin{pmatrix} \bar{y^2} \\ \bar{y} \end{pmatrix} = \begin{pmatrix} \bar{x^2} & \bar{x} \\ \bar{x} & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}$$

  - So what is the problem?

Line Estimation

- Least square minimization:
  - Line equation: $y = ax + b$
  - Error in fit: $\sum_i (y_i - ax_i - b)^2$
  - Solution:

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  - So what is the problem?
  - Minimizes vertical errors. Non-robust!

LSQ on Lasers

- Line model: $r_i \cos(\phi_i - \theta) = \rho$
- Error model: $d_i = r_i \cos(\phi_i - \theta) - \rho$
- Optimize: $\text{argmin}_{(\rho,\theta)} \sum_i (r_i \cos(\phi_i - \theta) - \rho)^2$
- Error model derived in Deriche et al. [1]
- Well suited for "clean-up" of Hough lines

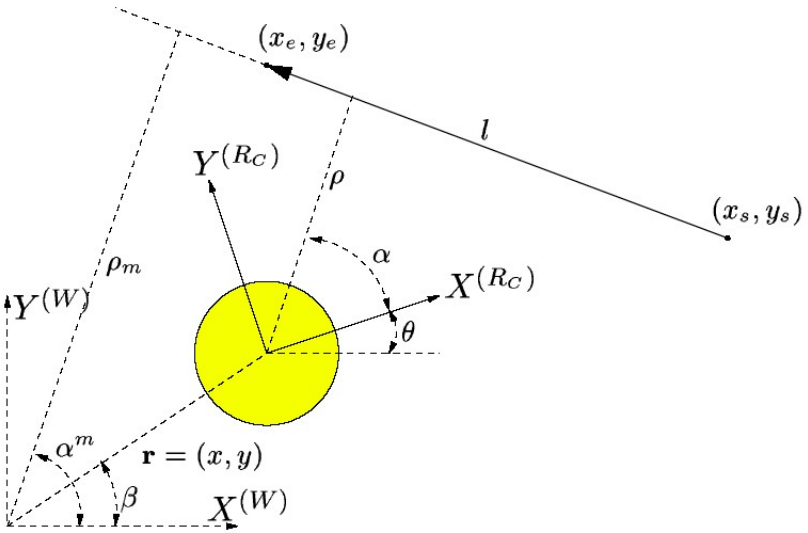
Total Least Squares

- Line equation: $ax + by + c = 0$
- Error in fit: $\sum_i (ax_i + by_i + c)^2$ where $a^2 + b^2 = 1$.
- Solution:

$$\begin{pmatrix} \bar{x^2} - \bar{x}\bar{x} & \bar{xy} - \bar{x}\bar{y} \\ \bar{xy} - \bar{x}\bar{y} & \bar{y^2} - \bar{y}\bar{y} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \mu \begin{pmatrix} a \\ b \end{pmatrix}$$

  where μ is a scale factor.
- $c = -a\bar{x} - b\bar{y}$

Line Representations



- The line representation is crucial
- Often a redundant model is adopted
- Line parameters vs end-points
- Important for fusion of segments.
- End-points are less stable

Sequential Adaptation

- In some cases one at a time estimation is more suitable
- Also known as gradient descent

$$\begin{aligned} \mathbf{w}^{(\tau+1)} &= \mathbf{w}^{(\tau)} - \eta \nabla E_n \\ &= \mathbf{w}^{(\tau)} - \eta(t_n - \mathbf{w}^{(\tau)T}\phi(x_n))\phi(x_n) \end{aligned}$$

- Knows as least-mean square (LMS). An issue is how to choose η?

Regularized Least Squares

- As seen in previous lecture sometime control of parameters might be useful.
- Consider the error function:

$$E_D(\mathbf{w}) + \lambda E_W(\mathbf{w})$$

- which generates

$$\frac{1}{2}\sum_{i=1}^{N}\{t_i - w^t\phi(x_i)\}^2 + \frac{\lambda}{2}\mathbf{w}^T\mathbf{w}$$

- which is minimized by

$$w = \left(\lambda I + \Phi^T\Phi\right)^{-1}\Phi^T\mathbf{t}$$

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Bayesian Linear Regression

- Define a conjugate prior over w

$$p(w) = N(w|m_0, S_0)$$

- given the likelihood function and regular from Bayesian analysis we can derive

$$p(w|t) = N(w|m_N, S_N)$$

- where

$$\begin{aligned} m_N &= S_N \left(S_0^{-1} m_0 + \beta \Phi^T t \right) \\ S_N^{-1} &= S_0^{-1} + \beta \Phi^T \Phi \end{aligned}$$

Bayesian Linear Regression (2)

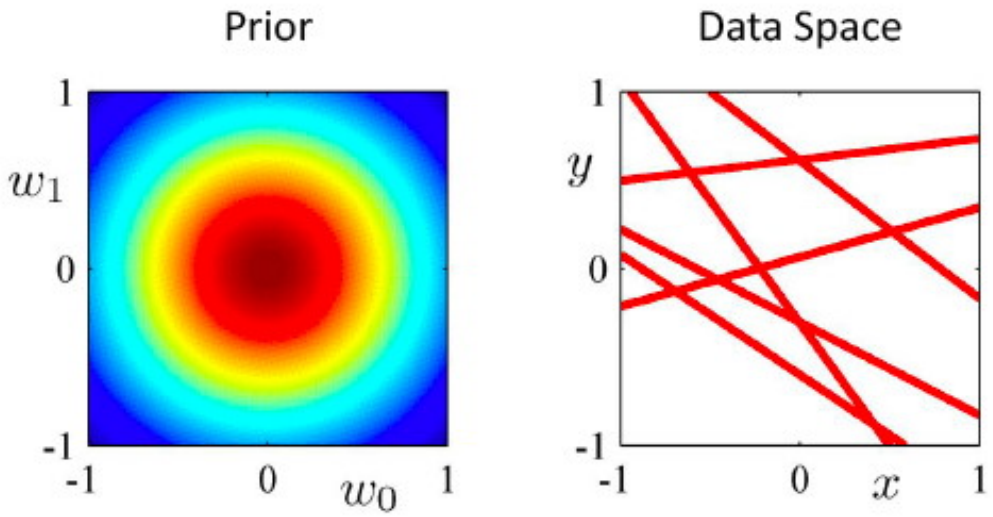
- A common choice is

$$p(w) = N(w|0, \alpha^{-1} I)$$

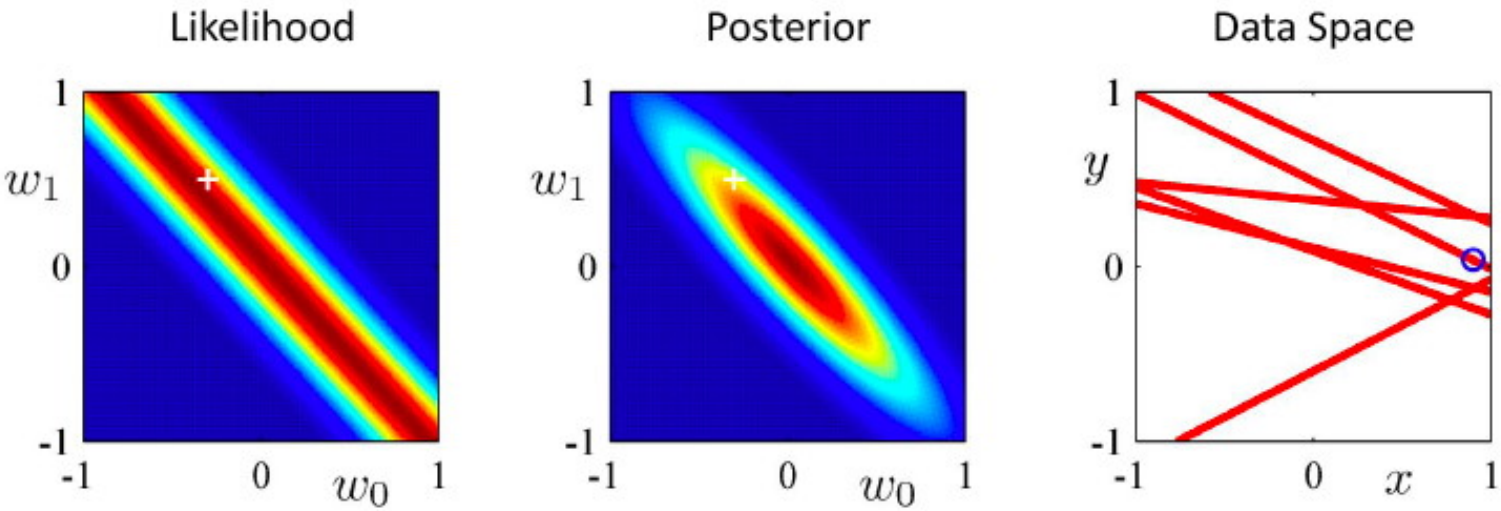
- So that

$$\begin{aligned} m_N &= \beta S_N \Phi^T t \\ S_N^{-1} &= \alpha I + \beta \Phi^T \Phi \end{aligned}$$

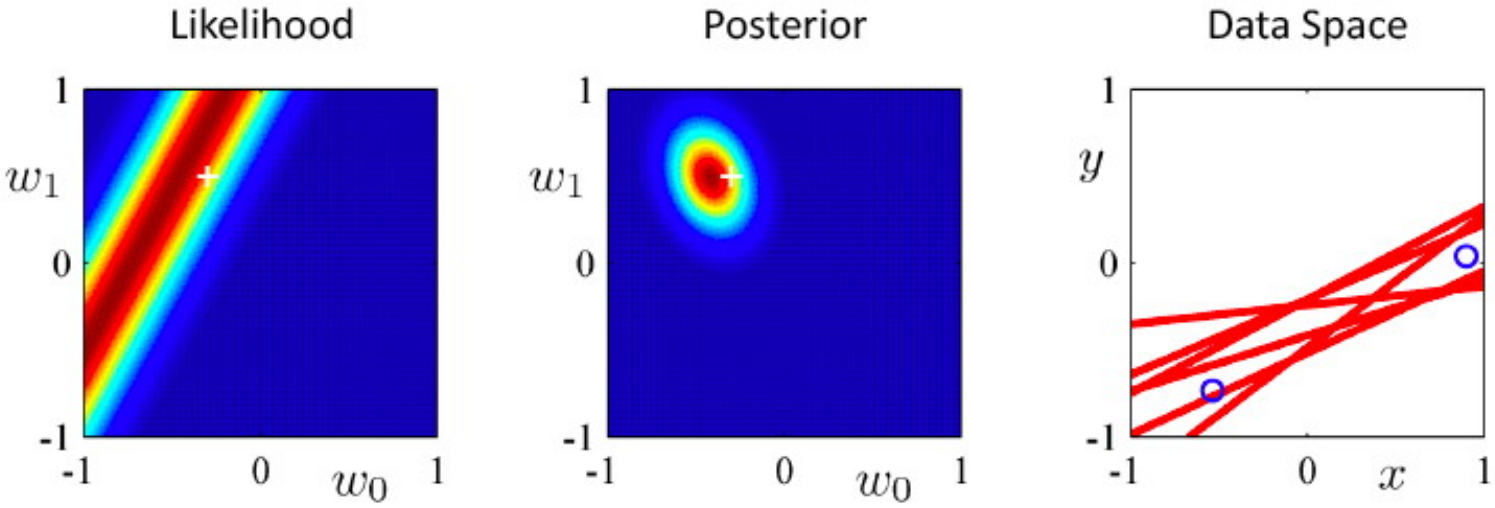
Example - No Data



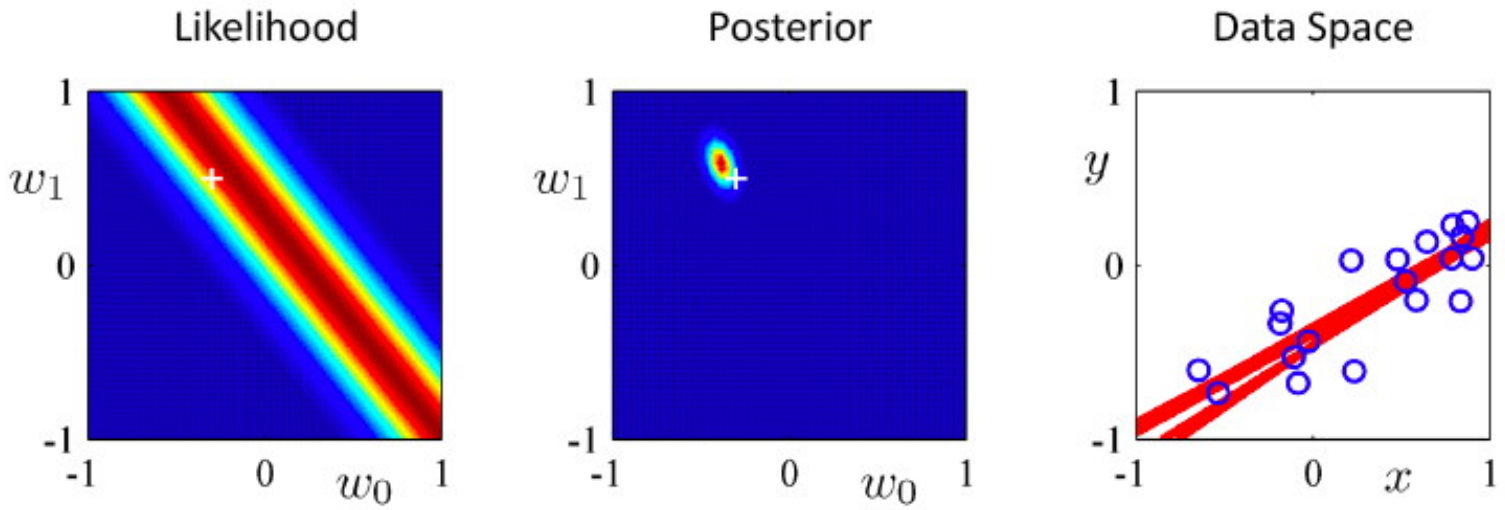
Example - 1 Data Point



Example - 2 Data Points



Example - 20 Data Points



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Bayesian Model Comparison

- How does one select an appropriate model?
- Assume for a minute we want to compare a set of models M_i, $i \in 1, ...L$ for a dataset D
- We could compute

$$p(M_i|D) \propto p(D|M_i)p(M_i)$$

- Bayes Factor: Ratio of evidence for two models

$$\frac{p(D|M_i)}{p(D|M_j)}$$

The mixture distribution approach

- We could use all the models:

$$p(t|x, D) = \sum_{i=1}^{L} p(t|x, M_i, D)p(M_i|D)$$

- Or simply go with the most probably/best model.

Model Evidence

- We can compute model evidence

$$p(D|M_i) = \int p(D|w, M_i)p(w|M_i)dw$$

- Allow computation of model fit based on parameter range

Evaluation of Parameters

- Evaluation of posterior over parameters

$$p(w|D, M_i) = \frac{P(D|w, M_i)p(w|M_i)}{P(D|M_i)}$$

- There is a need to understand how good is a model?

Model Comparison

- Consider evaluation of a model w. parameters w

$$p(D) = \int p(D|w)p(w)dw \approx p(D|w_{map})\frac{\sigma_{posterior}}{\sigma_{prior}}$$

- Then

$$\ln p(D) \approx \ln p(D|w_{map}) + \ln\left(\frac{\sigma_{posterior}}{\sigma_{prior}}\right)$$

Model Comparison as Kullback-Leibler

- From earlier we have comparison of distributions

$$KL = \int p(D|M_1) \ln \frac{p(D|M_1)}{p(D|M_2)} dD$$

- Enables comparison of two different models

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Regression Summary

- Brief intro to linear methods for estimation of models
- Prediction of values and models
  - Needed for adaptive selection of models (black-box/grey-box)
  - Evaluation of sensor models, ...
- Consideration of batch and recursive estimation methods
- Significant discussion of methods for evaluation of models and parameters.
- This far purely a discussion of linear models

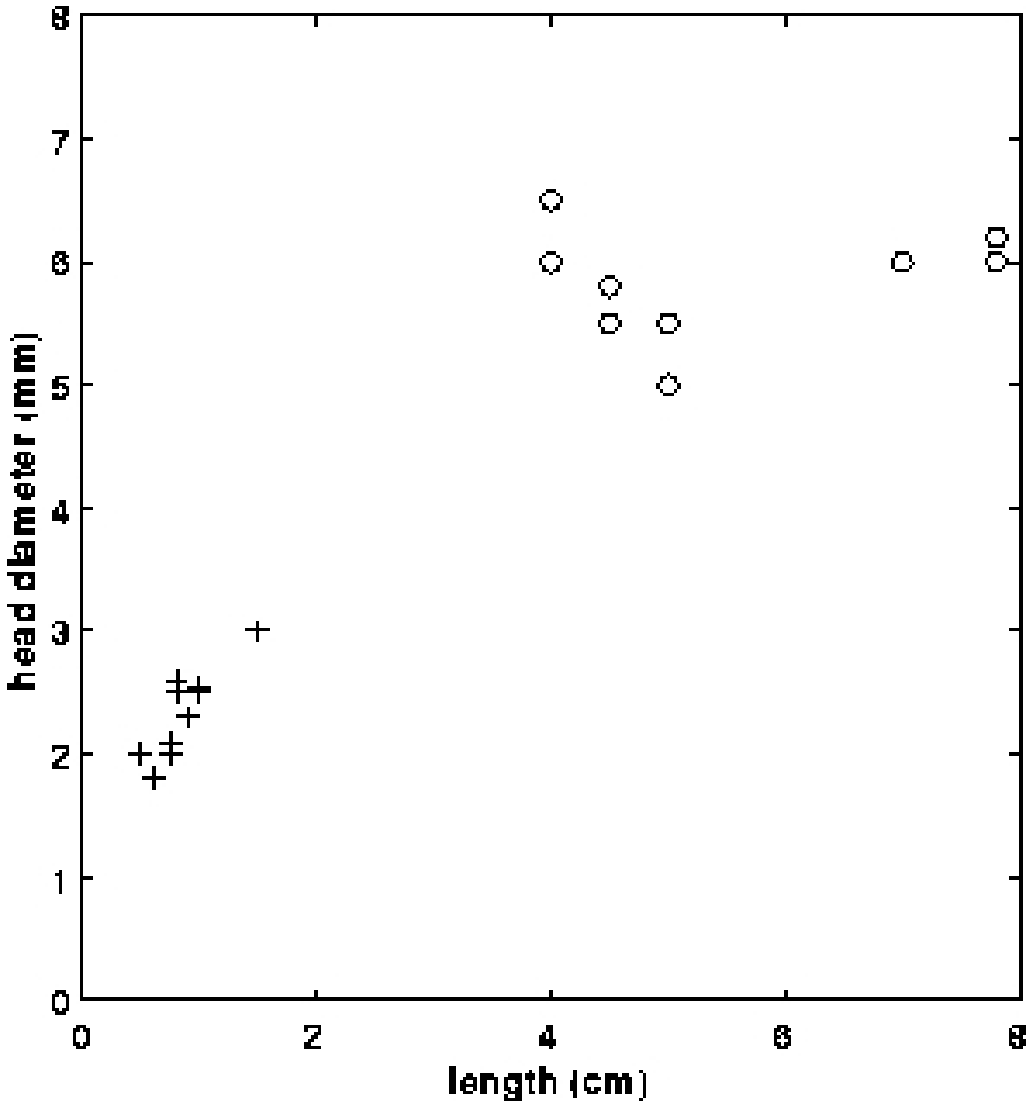
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Classification Introduction

- Linear classification of data
  - Basic pattern recognition
  - Separation of data: buy/sell
  - Segmentation of line data, ...

Simple Example - Bolts or Needles



Classification

- Given
  - An input vector: X
  - A set of classes: $c_i \in \mathcal{C}, \quad i = 1, \ldots, k$
- Mapping $m : X \rightarrow \mathcal{C}$
- Separation of space into decision regions
- Boundaries termed decision boundaries/surfaces

Basis Formulation

- It is a 1-of-K coding problem
- Target vector: $\mathbf{t} = (0, \ldots, 1, \ldots, 0)$
- Consideration of 3 different approaches
  1. Optimization of discriminant function
  2. Bayesian Formulation: $p(c_i|x)$
  3. Learning & Decision fusion

Code for experimentation

- There are data sets and sample code available
  - NETLAB: `http://www.ncrg.aston.ac.uk/netlab/index.php`
  - Kaggle: `https://www.kaggle.com`
  - Lots of good robotics data-sets too

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Discriminant Functions

- Objective: input vector x assigned to a class c_i
- Simple formulation:

$$y(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + w_0$$

- $\mathbf{w}$ is termed a weight vector
- w_0 is termed a bias
- Two class example: c_1 if $y(\mathbf{x}) \geq 0$ otherwise c_2

Basic Design

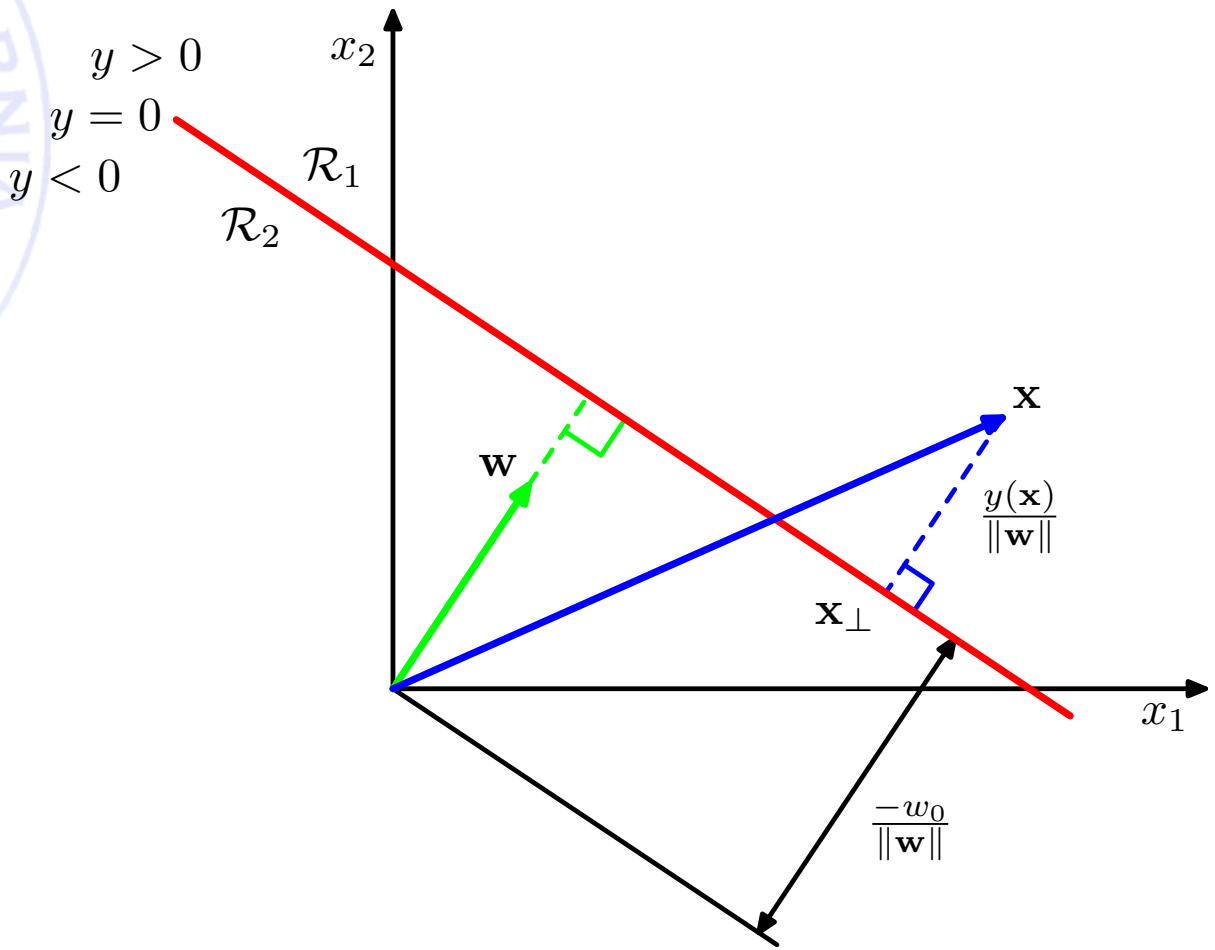
- Two points on decision surface $\mathbf{x}_a$ and $\mathbf{x}_b$
- $y(\mathbf{x}_a) = y(\mathbf{x}_b) = 0 \Rightarrow \mathbf{w}^T(\mathbf{x}_a - \mathbf{x}_b) = 0$
- $\mathbf{w}$ perpendicular to decision surface

$$\frac{\mathbf{w}^T\mathbf{x}}{||\mathbf{w}||} = -\frac{w_0}{||\mathbf{w}||}$$

- Define: $\tilde{w} = (w_0, \mathbf{w})$ and $\tilde{x} = (1, \mathbf{x})$ so that:

$$y(\mathbf{x}) = \tilde{w}^T\tilde{x}$$

Linear discriminant function



Multi Class Discrimination

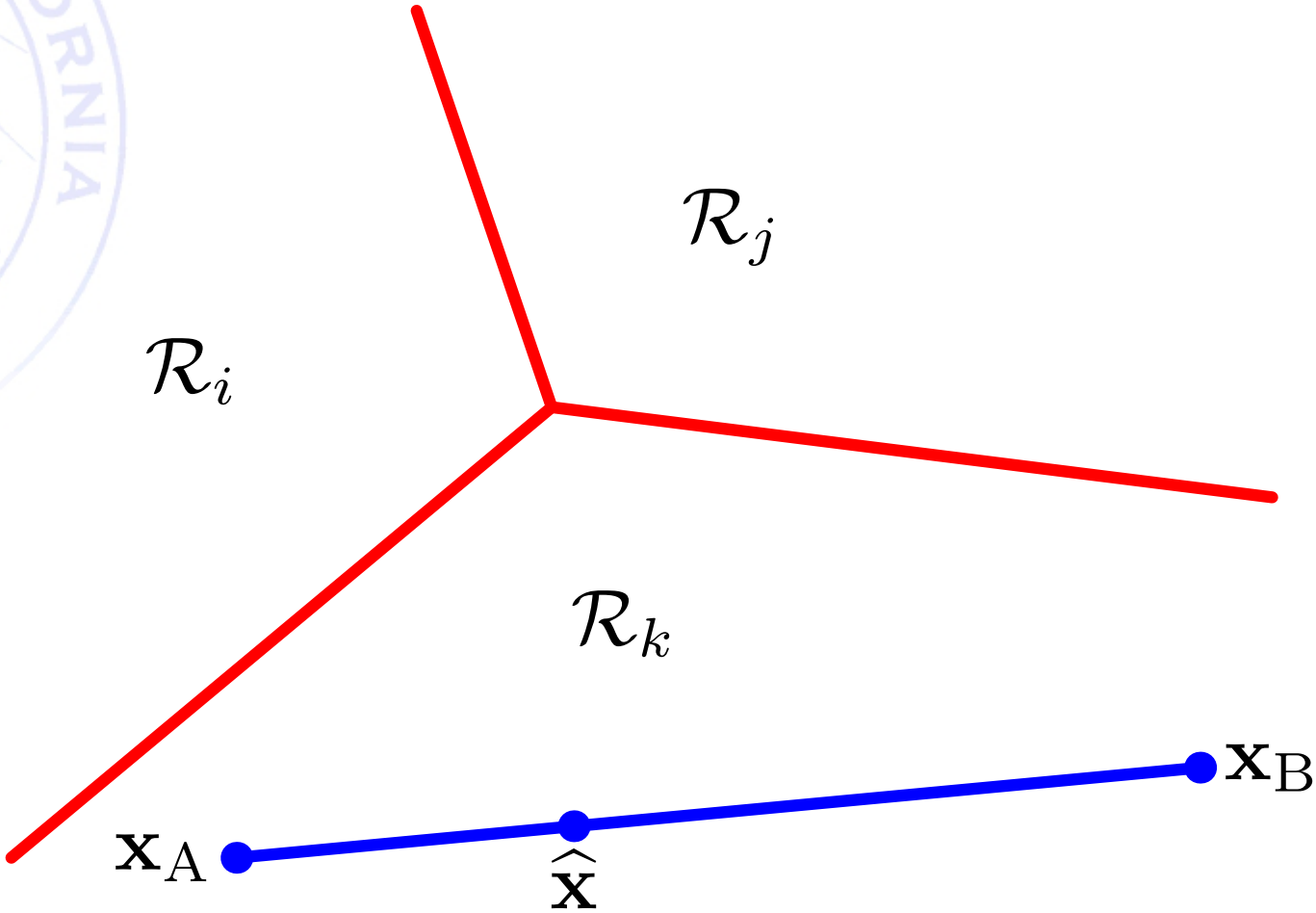
- Generation of multiple decision functions

$$y_k(\mathbf{x}) = \mathbf{w}_k^T \mathbf{x} + w_{k0}$$

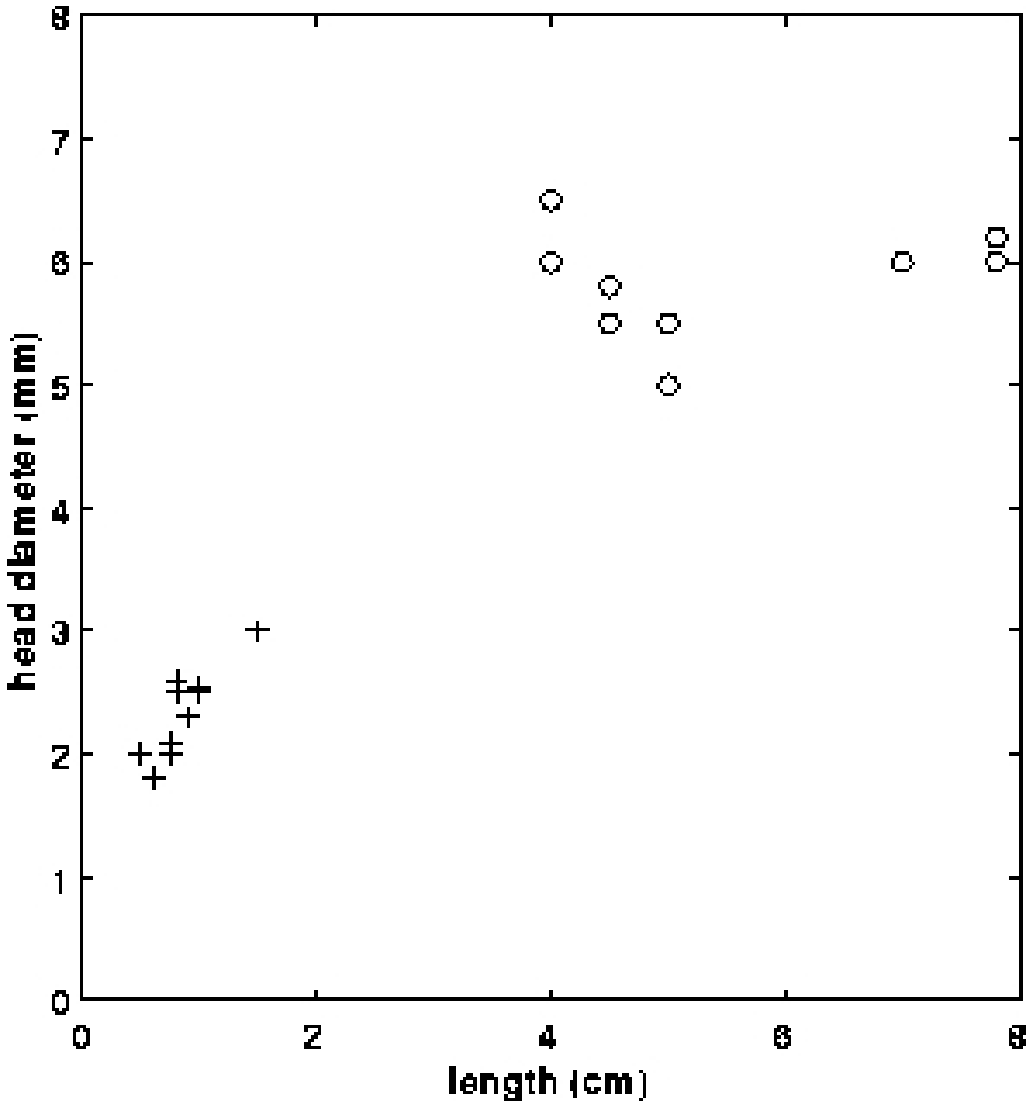
- Decision strategy

$$j = \arg \max_{i \in 1..k} y_i(\mathbf{x})$$

Multi-Class Decision Regions



Example - Bolts or Needles



Minimum distance classification

- Suppose we have computed the mean value for each of the classes
- $m_{needle} = [0.86, 2.34]^T$ and $m_{bolt} = [5.74, 5, 85]^T$
- We can then compute the minimum distance

$$d_j(x) = ||x - m_j||$$

- $\mathrm{a}rgmin_i d_i(x)$ is the best fit
- Decision functions can be derived

Bolts / Needle Decision Functions

Needle $d_{needle}(x) = 0.86x_1 + 2.34x_2 - 3.10$

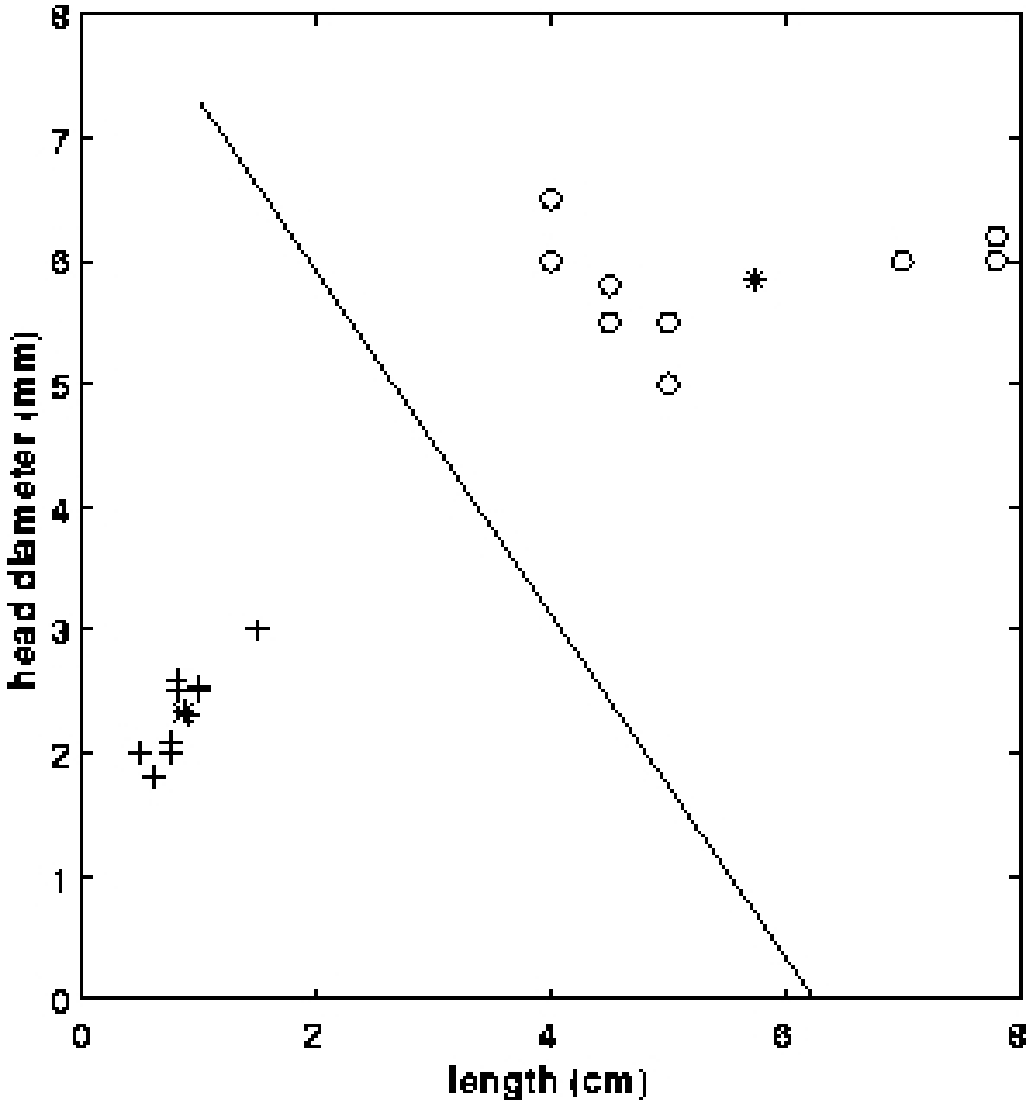
Bolt $d_{bolt}(x) = 5.74x_1 + 5.85x_2 - 33.59$

Decision boundary

$$d_i(x) - d_j(x) = 0$$

$$d_{needle/bolt}(x) = -4.88x_1 - 3.51x_2 + 30.49$$

Example decision surface



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Least Squares for Classification

- Just like we could do LSQ for regression we can perform an approximation to the classification vector $\mathcal{C}$
- Consider again

$$y_k(\mathbf{x}) = \mathbf{w}_k^T \mathbf{x} + w_{k0}$$

- Rewrite to

$$\mathbf{y}(\mathbf{x}) = \tilde{\mathbf{W}}^T \tilde{\mathbf{x}}$$

- Assuming we have a target vector $\mathbf{T}$

Least Squares for Classification

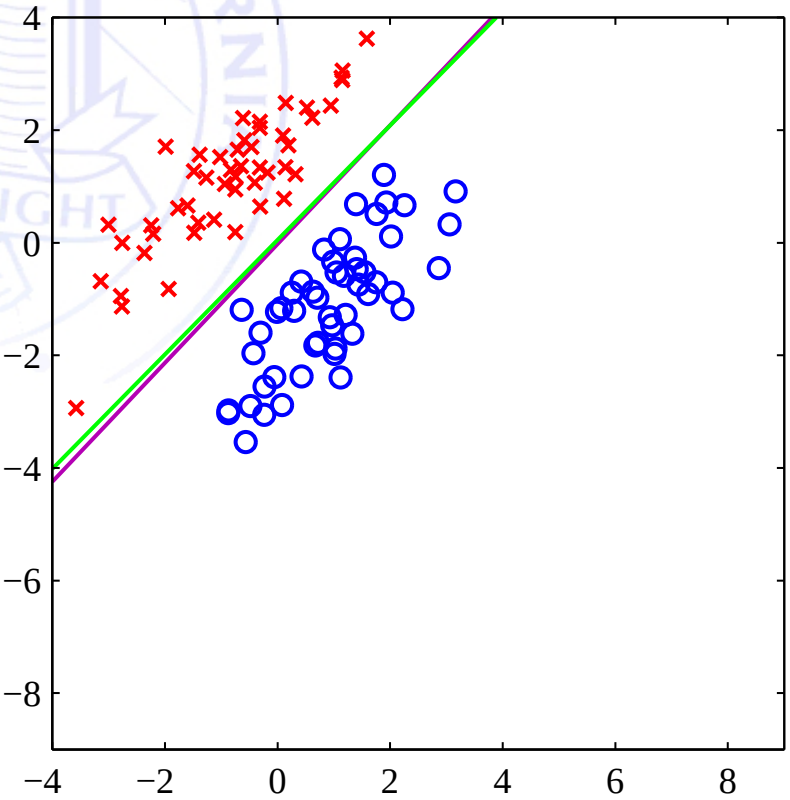
- The error is then:

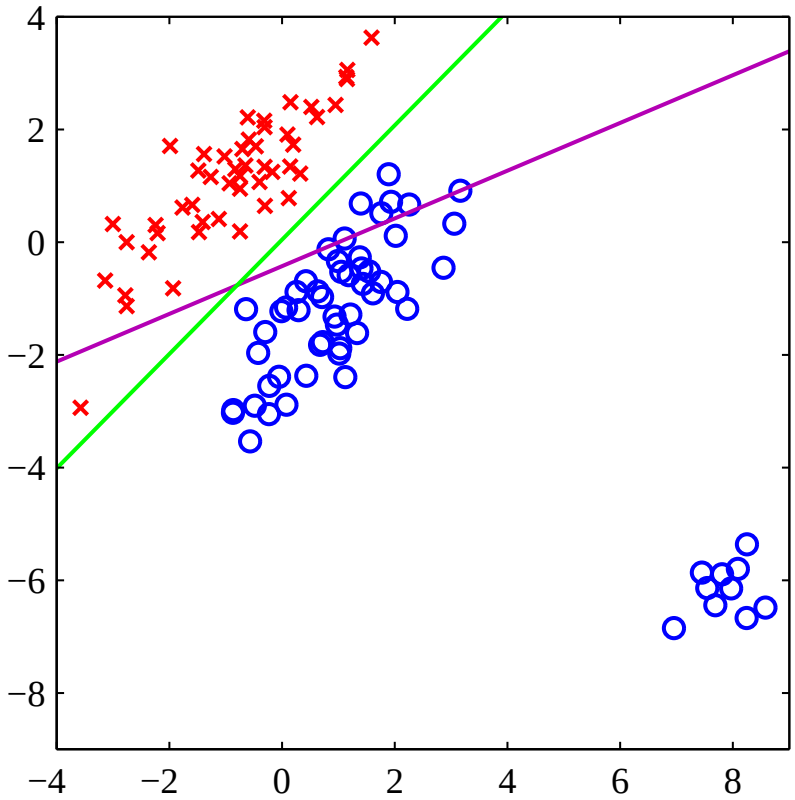
$$E_D(\tilde{\mathbf{W}}) = \frac{1}{2} Tr \left\{ (\tilde{\mathbf{X}}\tilde{\mathbf{W}} - \mathbf{T})^T (\tilde{\mathbf{X}}\tilde{\mathbf{W}} - \mathbf{T}) \right\}$$

- The solution is then

$$\tilde{\mathbf{W}} = \left(\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \right)^{-1} \tilde{\mathbf{X}}^T \mathbf{T}$$

LSQ and Outliers





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Fisher's linear discriminant

- Selection of a decision function that maximizes distance between classes
- Assume for a start

$$y = \mathbf{W}^T \mathbf{x}$$

- Compute m_1 and m_2

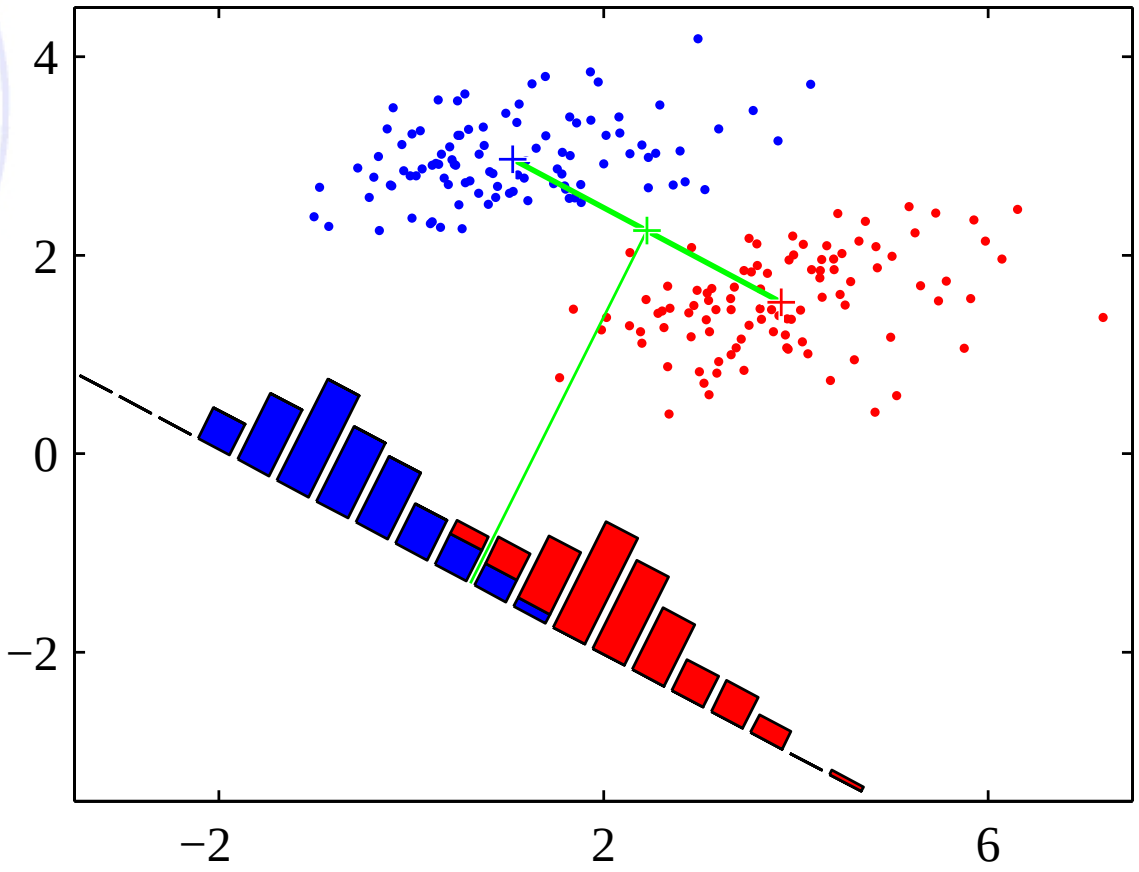
$$\mathbf{m}_1 = \frac{1}{N_1} \sum_{i \in C_1} \mathbf{x}_i \quad \mathbf{m}_2 = \frac{1}{N_2} \sum_{j \in C_2} \mathbf{x}_j$$

- Distance:

$$m_2 - m_1 = \mathbf{w}^T (\mathbf{m}_2 - \mathbf{m}_1)$$

- where $m_i = \mathbf{w}\mathbf{m}_i$

The sub-optimal solution



The Fisher criterion

- Consider the expression

$$J(w) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}$$

- where $\mathbf{S}_B$ is the between class covariance and $\mathbf{S}_W$ is the within class covariance, i.e.

$$\mathbf{S}_B = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^T$$

and

$$\mathbf{S}_W = \sum_{i=\mathcal{C}_1} (\mathbf{x}_i - \mathbf{m}_1)(\mathbf{x}_i - \mathbf{m}_1)^T + \sum_{i=\mathcal{C}_2} (\mathbf{x}_i - \mathbf{m}_2)(\mathbf{x}_i - \mathbf{m}_2)^T$$

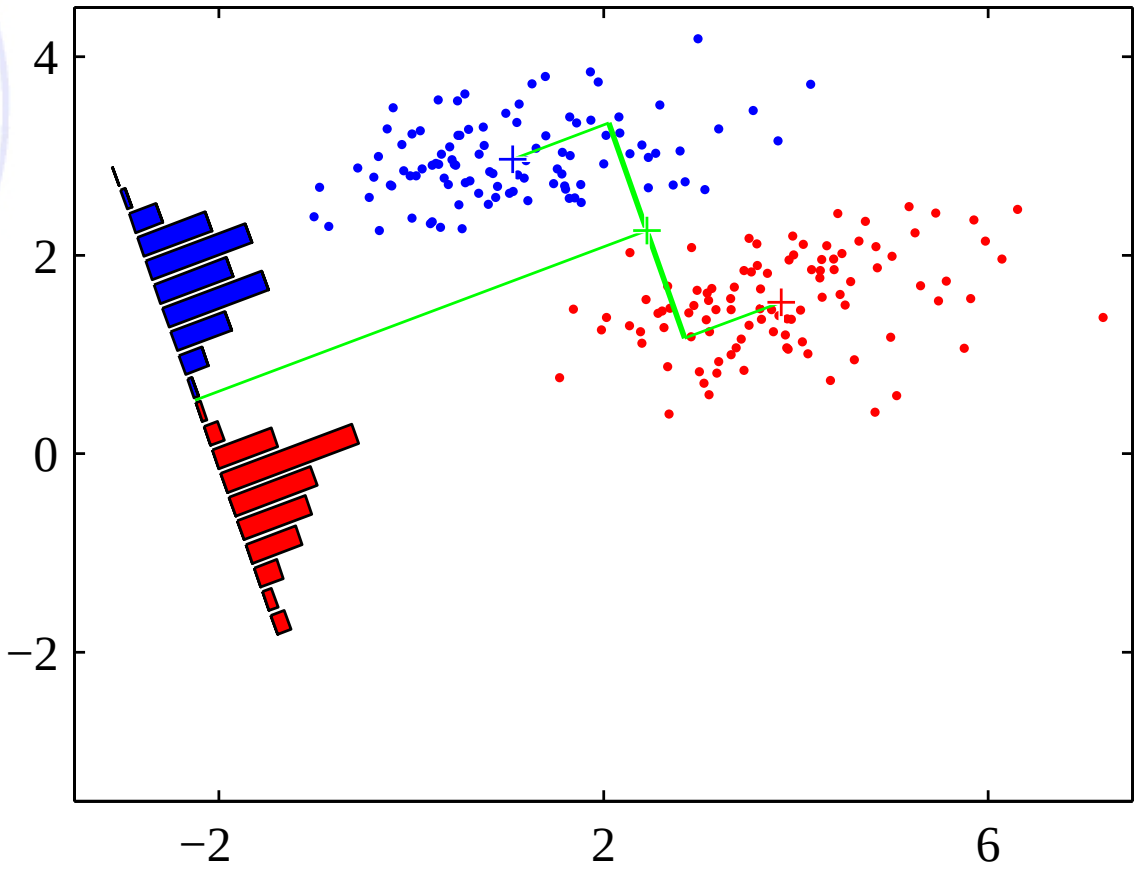
- Optimized when

$$(\mathbf{w}^T \mathbf{S}_B \mathbf{w}) \mathbf{S}_w \mathbf{w} = (\mathbf{w}^T \mathbf{S}_W \mathbf{w}) \mathbf{S}_B \mathbf{w}$$

or

$$\mathbf{w} \propto \mathbf{S}_w^{-1} (\mathbf{m}_2 - \mathbf{m}_1)$$

The Fisher result



Generalization to $N > 2$

- Define a stacked weight factor

$$\mathbf{y} = \mathbf{W}^T \mathbf{x}$$

- The within class covariance generalizes to

$$\mathbf{S}_w = \sum_{k=1}^{K} \mathbf{S}_k$$

- The between class covariance is

$$\mathbf{S}_B = \sum_{k=1}^{K} N_k (\mathbf{m}_k - \mathbf{m})(\mathbf{m}_k - \mathbf{m})^T$$

- It can be shown that $J(\mathbf{w})$ is optimized by the eigenvectors to the equation

$$S = \mathbf{S}_W^{-1} \mathbf{S}_B$$

Outline

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2. Preliminaries
3. Linear Basis Function Models
4. Bayesian Linear Regression
5. Bayesian Model Comparison
6. Regression Summary
7. Classification
8. Linear Discriminant Functions
9. LSQ for Classification
10. Fisher's Discriminant Method
11. Perceptrons
12. Summary

Perceptron Algorithm

- Developed by Rosenblatt (1958) [2]
- Formed an important basis for neural networks
- Use a non-linear transformation $\phi(x)$
- Construct a decision function

$$y(\mathbf{x}) = f\left(\mathbf{w}^T \phi(\mathbf{x})\right)$$

- where

$$f(a) = \begin{cases} +1, & a \geq 0 \\ -1, & a < 0 \end{cases}$$

The perceptron criterion

- Normally we want

$$w^t \phi(x_n) > 0$$

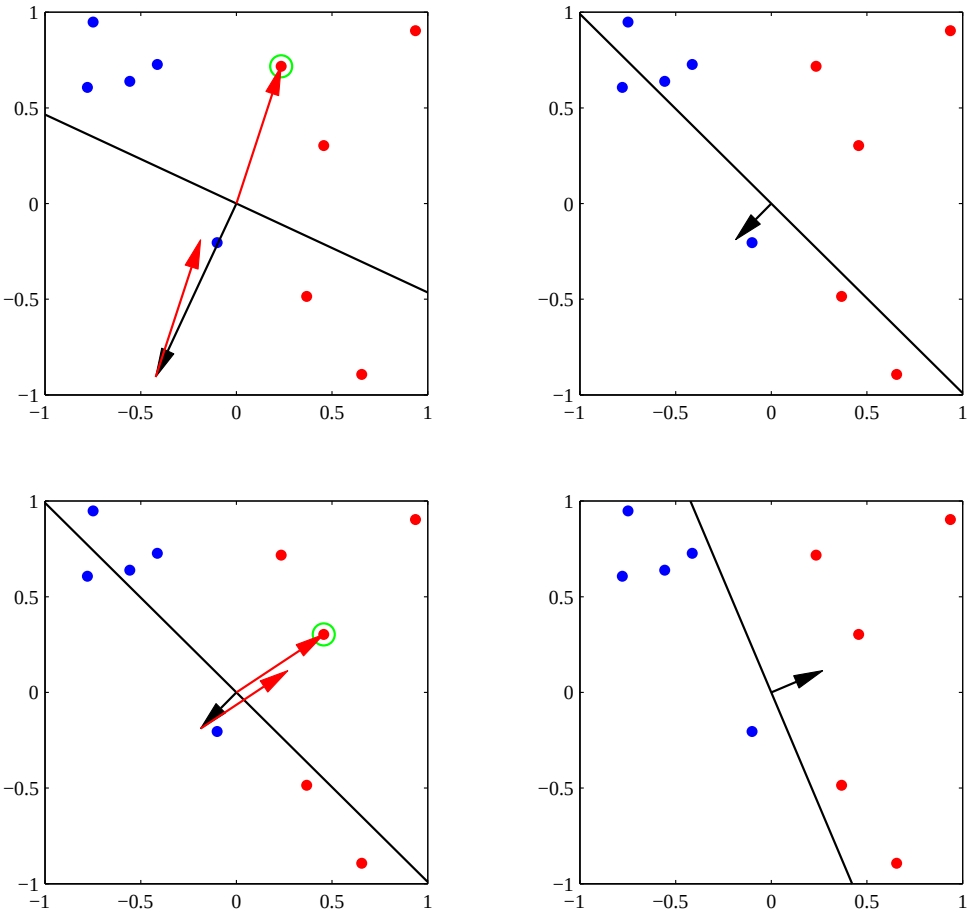
- Given the target vector definition

$$E_p(\mathbf{w}) = -\sum_{n \ in \mathcal{M}} \mathbf{w}^T \phi_n t_n$$

- Where $\mathcal{M}$ represents all the mis-classified samples
- We can make this a gradient descent as seen in last lecture

$$\mathbf{w}^{(\tau+1)} = \mathbf{w}^{(\tau)} - \eta \nabla E_P(\mathbf{w}) = \mathbf{w}^{(\tau)} + \eta \phi_n t_n$$

Perceptron learning example



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Classification Summary

- Basics for discrimination / classification
- Obviously not all problems are linear
- Optimization of the distance/overlap between classes
  - Minimizing the probability of error classification
- Basic formulation as an optimization problem
- How to optimize between cluster distance? Covariance Weighted
- Basic recursive formulation
- Could we make it more robust?

Summary

- Data models are anchored in pure data driven or model based evaluation
- How can we use models to interpret data and extrapolate beyond the basic data?
- Covered basic models for regression and classification.

R. Deriche, R. Vaillant, and O. Faugeras, "From Noisy Edges Points to 3D Reconstruction of a Scene : A Robust Approach and Its Uncertainty Analysis," in *In Theory and Applications of Image Analysis*, vol. 2, pp. 71–79, World Scientific, 1992.

F. Rosenblatt, "The perceptron: a probabilistic model for information storage and organization in the brain," *Psychological Review*, vol. 65, pp. 386–408, 1958.